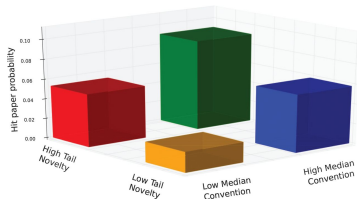


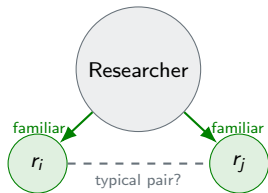
Knowledge Recombination is Shaped by Retrieval Structure



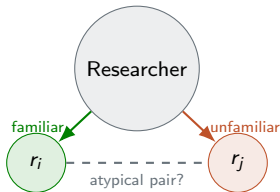
Atypical combinations, Uzzi et al. (2013)

Rarer combinations may lead to higher impact.

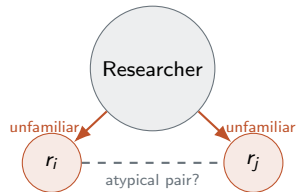
Local search



Pivoting (anchored exploration)



Leaping (unanchored exploration)



Two Dimensions of Knowledge Recombination

Invention (Innovation) is a **process of recombinant search** (*Fleming, 2001*).

- **Search process:** search/retrieve existing knowledge components.
 - e.g., **physicists** are more likely to retrieve **physics** than **biology**.
 - Conditional probability $\Pr(k_i | H_a)$.
- **Combination process:** combine knowledge components into new configurations.
 - e.g., **physics** and **astrophysics** are more likely to be combined than **physics** and **biology**.
 - Joint probability $\Pr(k_i, k_j)$.

Component unfamiliarity (Local distance, retrieval cost)	Combination atypicality (Global uncommonness, rarity)
How likely is a researcher to reach r_i ?	How unusual is the pair (r_i, r_j) in science?
Exploitation / exploration	Typical / atypical
$S(r H_a) = -\log \Pr(r H_a)$	$A(r_i, r_j) = -\text{npmi}(r_i, r_j)$

Measurement: Trajectory-conditioned Component Unfamiliarity

- **Component unfamiliarity**: how unexpected a knowledge component (cited journal) is given a researcher's prior retrieval trajectory.
- A **sequential model** $\mathcal{M}$ predicts journal r a researcher is likely to cite in the next paper, conditional on their prior publication history H_a :

$$\Pr(r|H_a) = \mathcal{M}(H_a)$$

- For each observed cited journal r_i , **retrieval surprisal** is defined as:

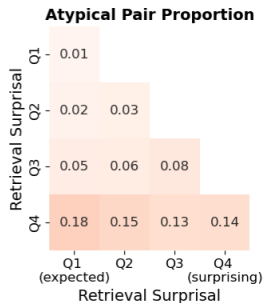
$$S(r_i|H_a) = -\log \Pr(r_i|H_a)$$

- Low value means the component is locally accessible (familiar) to the researcher; high value means the component is distant (unfamiliar) to the researcher.

Results: Anchored Atypicality

- Pair-level structure

Local search low-low	Pivot low-high
Pivot high-low	Leap high-high



- Atypicality is highest in asymmetric retrieval profiles, where one component is accessible and the other is distant.

- Paper-level consequences

Pair type	Top-5% citation	Disruption index
Anchored atypical	4.367***	-0.003***
Unanchored atypical	2.608***	0.002***
Anchored typical	1.249***	-0.002***
Unanchored typical	0 (baseline)	0 (baseline)

- Anchored atypical pairs are more likely to be highly cited than unanchored atypical pairs.
- Anchored atypical pairs consolidate, while unanchored atypical pairs disrupt.

- I leveraged the proposed component-level measure $S(r|H_a)$ to analyze how search process shapes the combination process.
- The same framework can be used as a diagnostic tool for research trajectories.
- This will enable trajectory-aware analysis of research careers and collaborations.
 - How do researchers' careers and collaborations evolve?
 - When do researchers successfully pivot? (When anchored with familiar components)
 - When do collaborators create new accessible frontiers?

saeyoung.choi@snu.ac.kr

